1-week VaR estimation: DAX & SPX

Contents

1. Introduction

Classical VaR often relies on joint normality and linear correlation, which can be unreliable for financial returns due to skewness, heavy tails, volatility clustering, and asymmetric dependence in extremes. Copulas offer a principled alternative: they model the dependence structure separately from the marginal distributions, allowing more realistic tail behaviour and flexible joint modelling.

A key requirement for static copula estimation is that the copula inputs are approximately i.i.d. over time. Concretely, after transforming each marginal to uniforms, the pseudo-observations $(u_{\text{SPX},t}, u_{\text{DAX},t})$ should behave like independent draws from a time-invariant bivariate distribution. Raw equity returns often violate this because volatility clustering and other dynamics induce serial dependence in the conditional distribution.

Our workflow is therefore: (i) exploratory diagnostics to check for heavy tails, skewness, autocorrelation and volatility clustering; (ii) fit ARMA-GARCH models to each index to capture conditional mean dynamics and conditional heteroskedasticity, yielding standardised innovations that are closer to i.i.d.; (iii) apply the probability integral transform (PIT) to these innovations to obtain uniform pseudo-observations; (iv) fit and select a copula on $(u_{\text{SPX},t}, u_{\text{DAX},t})$; and (v) estimate 1-week 95% and 99% VaR for the weekly rebalanced portfolio via Monte Carlo simulation under the fitted marginal–copula model.

2. Main Task

2.1 Initialisation

```r
library(tseries)
library(fGarch)
library(KScorrect)
library(VineCopula)
library(goftest)
library(rugarch)


# Data Importing and Preparation
spx_w <- read.csv("^spx_w.csv")
dax_w <- read.csv("^dax_w.csv")


# Extract close prices
spx_close<-spx_w$Close
dax_close<-dax_w$Close


# Convert Date column
spx_w$Date <- as.Date(spx_w$Date)
dax_w$Date <- as.Date(dax_w$Date)

# Keep only Date + Close
spx_close_df <- spx_w[, c("Date", "Close")]
dax_close_df <- dax_w[, c("Date", "Close")]

# Set start and end date
START_DATE <- as.Date("2000-01-02")
END_DATE   <- as.Date("2025-12-28")
```

```r
dax_close_df <- dax_close_df[dax_close_df$Date >= START_DATE & dax_close_df$Date <= END_DATE, ]
spx_close_df <- spx_close_df[spx_close_df$Date >= START_DATE & spx_close_df$Date <= END_DATE, ]

# Rename close columns so they don't collide
names(spx_close_df)[2] <- "SPX"
names(dax_close_df)[2] <- "DAX"


prices <- merge(spx_close_df, dax_close_df, by = "Date", all = FALSE)
prices <- prices[complete.cases(prices), ]      # Drop any remaining NAs
prices <- prices[order(prices$Date), ]          # Sort by date
```

2.2 Log-returns and portfolio definition

```r
# Creation of the log-returns of the indices
rSPX <- diff(log(prices$SPX))
rDAX <- diff(log(prices$DAX))

returns <- data.frame(
  Date = prices$Date[-1],
  rSPX = rSPX,
  rDAX = rDAX
)

# equally weighted portfolio (log-return approximation)
returns$rP <- 0.5 * returns$rSPX + 0.5 * returns$rDAX

# equally weighted portfolio with weekly rebalancing (consistent with your simulation definition)
R1 <- exp(returns$rSPX) - 1
R2 <- exp(returns$rDAX) - 1
returns$rP_rebal <- log(1 + 0.5*R1 + 0.5*R2)
```

Pricing data was obtained from Stooq (Stooq n.d.). Let $p_{j,t}$ be the weekly closing price for index j $\in$ {SPX, DAX} at week t. Weekly log-return:

$$r_{j,t} \;=\; \log(P_{j,t}) - \log(P_{j,t-1}) \;=\; \log\!\left(\frac{P_{j,t}}{P_{j,t-1}}\right)$$

To construct an equally weighted Weekly portfolio log-return:

$$R_{p,t} = \frac{1}{2}\,(e^{r_{\mathrm{SPX},t}} - 1) + \frac{1}{2}\,(e^{r_{\mathrm{DAX},t}} - 1), \qquad r_{p,t} = \log\bigl(1 + R_{p,t}\bigr)\,.$$

We define the portfolio loss fraction for 1-week (relative to the current value V_0) as

$$L_t \;=\; -\,r_{p,t}.$$

Thus, VaR_c at confidence level $c \in \{0.95, 0.99\}$ is the c-quantile of L:

$$\mathrm{VaR}_c \;=\; \mathrm{Quantile}_c(L).$$

2.3 Exploratory analysis

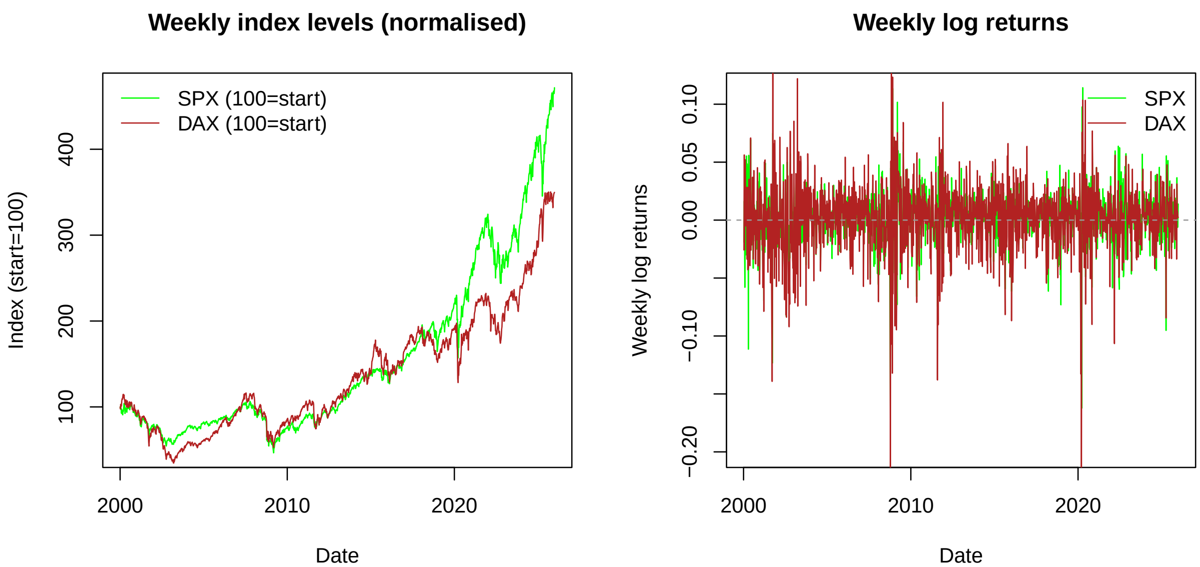

Figure 1: Weekly normalised index levels for the SPX and DAX (left) and their weekly log returns (right) from 2000–2025.

To compare long-run behaviour on a common scale, we normalise each index to start at 100 and plot weekly index levels (Figure 1). Both SPX and DAX trend upward over the sample with pronounced drawdowns during stress periods (notably 2008–2009 and 2020). The corresponding weekly log-return series fluctuates around zero and shows clear volatility clustering (bursts of large moves followed by calmer periods), motivating a conditional heteroskedastic model in each marginal series.

Index	Mean	SD	Skew	Excess Kurtosis	JB	JB_p
SPX	0.0011438784	0.02483206	-0.8709850	7.071785	3009.296	0
DAX	0.000923694	0.03083369	-0.8803177	6.782680	2785.880	0

Table 1: Summary statistics and Jarque-Bera (JB) tests for weekly log-returns of SPX and DAX.

Histograms and normal Q–Q plots (Figure 2) show strong departures from normality in both tails, with particularly large deviations in the lower tail. This visual evidence is consistent with the numerical summaries: both return series have pronounced negative skewness (approximately -0.87 to -0.88) and substantial excess kurtosis (approximately 6.8 to 7.1), indicating asymmetric heavy lower tails where extreme negative returns occur more frequently and with larger magnitude than extreme positive returns.

Normality is assessed formally using the Jarque–Bera (JB) where small p-values provide evidence against normality. The JB test strongly rejects normality for both indices (SPX: JB $\approx$ 3009; DAX: JB $\approx$ 2786; both with $p < 2.2 \times 10^{-16}$) (Table 1). Overall, the plots and tests motivate heavy-tailed, potentially skewed marginal innovations and dependence modelling beyond using simple linear correlation.

The ACF/PACF plots (Figure 3) show that most return autocorrelations lie within the 95% bounds, so any mean dependence is weak and, if present, likely low order. By contrast, the ACF of squared returns is clearly positive at short lags for both indices, indicating volatility clustering and motivating a GARCH-type variance model. Combined with the large excess kurtosis (Table 1), this points to an ARMA–GARCH specification with low-order mean dynamics and heavy-tailed innovations; we assess the ARMA order formally in the next section.

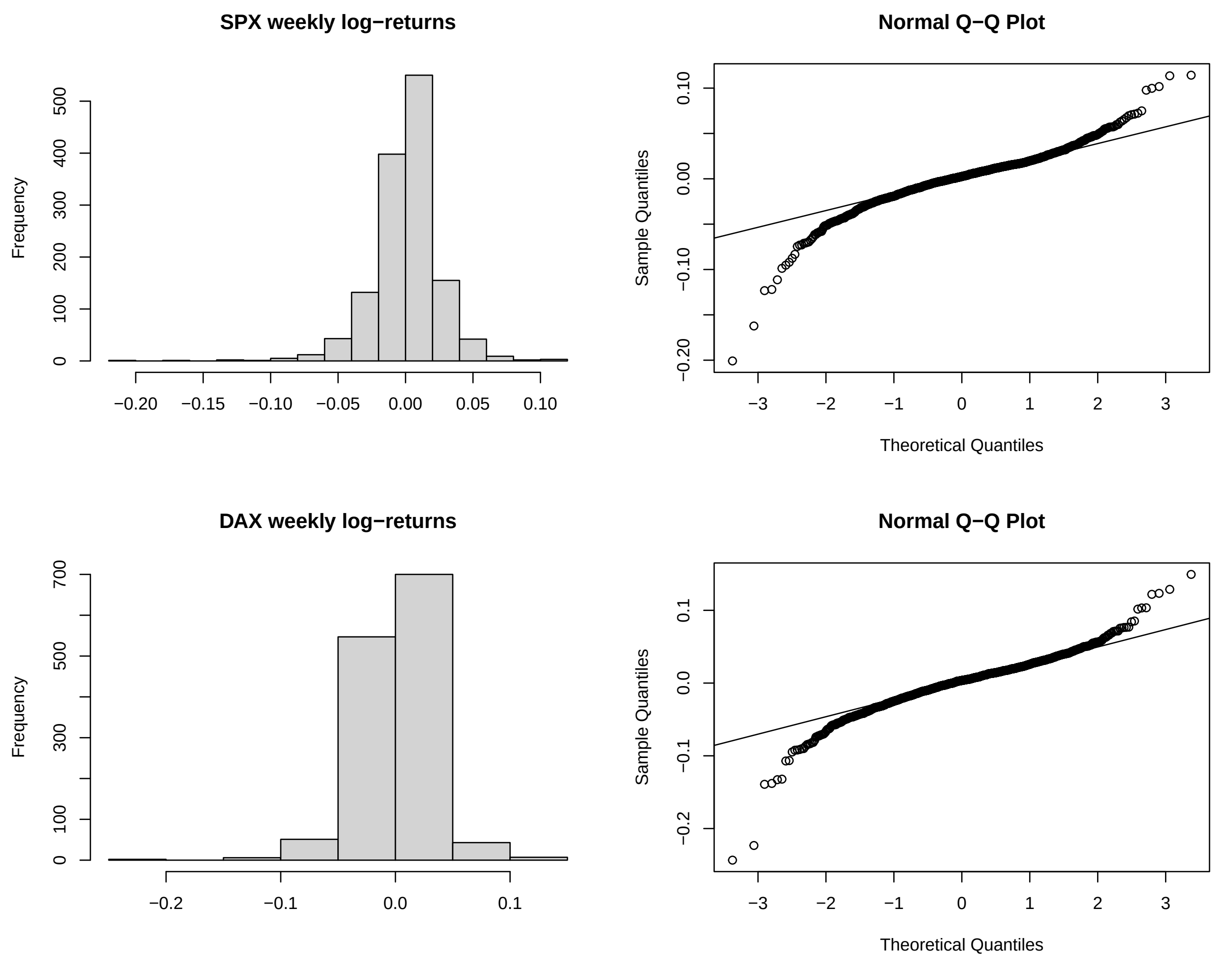

Figure 2: Histograms and normal Q–Q plots of weekly log-returns for the SPX and DAX.

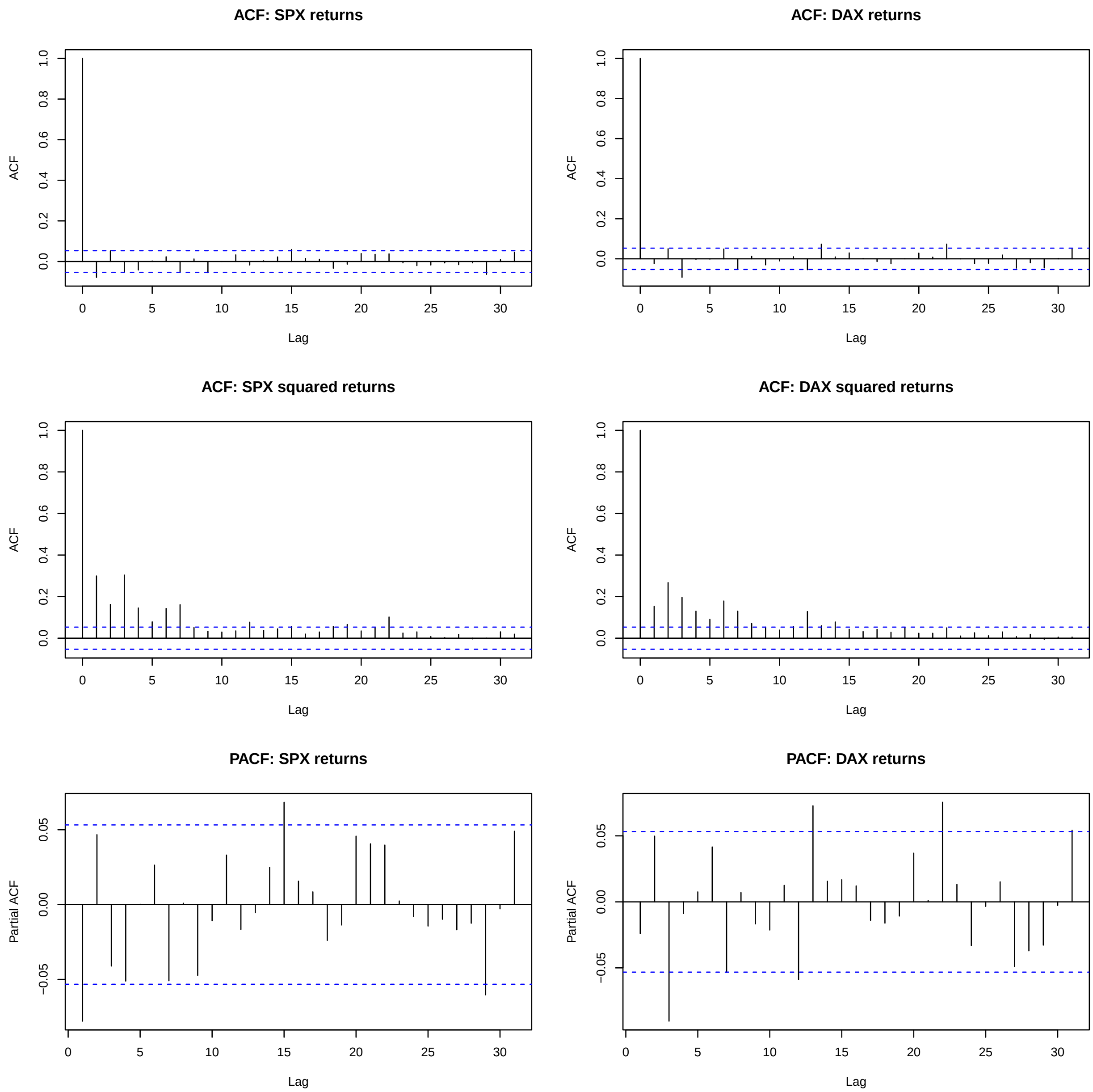

Figure 3: ACF and PACF plots for SPX and DAX weekly log-returns and squared returns.

2.4 Conditional mean model selection

Following the Box–Jenkins approach, we compare low-order AR(p) specifications with p=0,1,2,3 as return ACF/PACF indicated at most weak short-lag dependence and no stable structure consistent with higher-order mean dynamics:

$$r_{j,t} = \mu_{j,t} + \varepsilon_{j,t}, \qquad \mu_{j,t} = c_j + \sum_{i=1}^{p} \phi_{j,i}\, r_{j,t-i} + \sum_{k=1}^{q} \theta_{j,k}\, \varepsilon_{j,t-k}.$$

Candidate models are estimated via maximum likelihood and compared using AIC/BIC (prioritising BIC to favour parsimony). The selected specification is then checked using Ljung–Box tests on the standardised residuals, which test the joint null hypothesis of no autocorrelation up to a chosen lag (so rejection would indicate remaining serial dependence and an inadequate mean specification).

p	AIC	BIC
0	-6171.360	-6160.935
1	-6177.608	-6161.971
2	-6178.569	-6157.720
3	-6178.859	-6152.798

Table 2: AIC and BIC values for AR(p) models fitted to SPX returns.

Model	Q statistic	df	p-value
ARMA(1,0)	16.261	10	0.09239

Table 3: Ljung–Box test on fitted standardised residuals for selected SPX mean model.

For SPX, BIC is minimised at $p = 1$ (Table 2), and the LB tests on fitted residuals across 10 lags indicated no remaining linear autocorrelation at standard lags (p>0.05) (Table 3). Therefore we proceed with an ARMA(1,0) mean specification.

p	AIC	BIC
0	-5584.287	-5573.862
1	-5583.067	-5567.430
2	-5584.443	-5563.594
3	-5593.600	-5567.538

Table 4: AIC and BIC values for AR(p) models fitted to DAX returns.

For DAX, the BIC-selected ARMA$(0, 0)$ specification (Table 4) produced a significant Ljung–Box statistic on the fitted (standardised) residuals (p-value < 0.05) (Table 5), indicating remaining serial dependence in the conditional mean. We therefore evaluated higher-order mean specifications, and retained the ARMA$(3, 0)$ model as the lowest-order choice for which the Ljung–Box test failed to reject the null of no autocorrelation at the 5% level (p-value > 0.05) (Table 5).

Therefore the final conditional mean models are ARMA(1,0) for SPX and ARMA(3,0) for DAX.

Model	Q statistic	df	p-value
ARMA(0,0)	24.5770	10	0.006207
ARMA(1,0)	22.7740	10	0.01161
ARMA(2,0)	19.2640	10	0.03704
ARMA(3,0)	7.7231	10	0.6559

Table 5: Ljung–Box tests on fitted standardised residuals for DAX mean model candidates.

2.5 Conditional variance model selection

Series / fitted model	Q statistic	df	p-value
SPX (mean model residuals)^2	330.31	10	$< 2.2 \times 10^{-16}$
DAX (mean model residuals)^2	267.80	10	$< 2.2 \times 10^{-16}$

Table 6: Ljung–Box tests on squared residuals (testing for remaining ARCH clustering).

The Ljung–Box test applied to the squared residuals from the fitted ARMA mean models is significant for both series (p-value < 0.05), corroborating the positive autocorrelation observed in the ACF of squared mean-model residuals (Table 6). This provides formal evidence of ARCH-type dependence (volatility clustering) and motivates a GARCH specification for the conditional variance:

$$\varepsilon_{j,t} = \sigma_{j,t} z_{j,t}, \qquad \sigma^2_{j,t} = \omega_j + \alpha_j \varepsilon^2_{j,t-1} + \beta_j \sigma^2_{j,t-1},$$

where $z_{j,t}$ are i.i.d. innovations.

Holding the conditional mean fixed at the selected ARMA orders (SPX: ARMA$(1,0)$; DAX: ARMA$(3,0)$), we compare low-order GARCH(p,q) specifications with $(p,q) \in \{(1,1),(1,2),(2,1),(2,2)\}$ via maximum likelihood. Since both return series exhibit heavy tails, these candidate GARCH models are initially estimated under Student-t innovations to avoid variance underestimation under Gaussian errors. Models are compared using AIC/BIC (prioritising BIC to favour parsimony).

Model	AIC	BIC	SIC	HQIC
g11	-4.909832	-4.886769	-4.909871	-4.901197
g12	-4.908842	-4.881935	-4.908895	-4.898768
g21	-4.908750	-4.881843	-4.908803	-4.898675
g22	-4.907374	-4.876623	-4.907443	-4.895860

Table 7: Information criteria for candidate GARCH models fitted to SPX returns.

Model	AIC	BIC	SIC	HQIC
g11	-4.421803	-4.391052	-4.421872	-4.410289
g12	-4.420576	-4.385981	-4.420664	-4.407623
g21	-4.420293	-4.385698	-4.420380	-4.407340
g22	-4.419101	-4.380663	-4.419209	-4.404709

Table 8: Information criteria for candidate GARCH models fitted to DAX returns.

For DAX, the information criteria agree: GARCH$(1,1)$ attains the lowest AIC and BIC among the candidate orders, so we retain GARCH$(1,1)$ for the conditional variance (Table 8).

For SPX, the information criteria are very close across GARCH$(1,1)$, GARCH$(1,2)$, GARCH$(2,1)$ and GARCH$(2,2)$: AIC is minimised by GARCH$(1,1)$, while BIC is marginally lower for GARCH$(2,2)$ (Table

7). Given the negligible differences and the preference for a parsimonious volatility specification, we select GARCH$(1, 1)$ for SPX as well.

Conditional on the selected ARMA–GARCH orders, we now compare alternative innovation distributions within this structure and select the final conditional distribution using likelihood-based criteria (Table 9 and Table 10).

Distribution	AIC	BIC
Normal	-4.854768	-4.835549
Student-t	-4.909832	-4.886769
Skewed Student-t	-4.941944	-4.915037
Skewed Normal	-4.910392	-4.887328
Skewed Generalised Error Distribution	-4.935027	-4.908120
Generalised Error Distribution	-4.896581	-4.873517

Table 9: Information criteria for alternative innovation distributions in the ARMA–GARCH model for SPX returns.

Distribution	AIC	BIC
Normal	-4.354826	-4.327918
Student-t	-4.421803	-4.391052
Skewed Student-t	-4.459454	-4.424859
Skewed Normal	-4.417081	-4.386330
Skewed Generalised Error Distribution	-4.447922	-4.413328
Generalised Error Distribution	-4.402475	-4.371724

Table 10: Information criteria for alternative innovation distributions in the ARMA–GARCH model for DAX returns.

Across both indices, the skewed Student-t innovation attains the lowest information criteria, so it is retained as the best-fitting conditional distribution (Table 9 and Table 10). This aligns with the sample summary statistics (excess kurtosis and mild skewness), implying heavy-tailed and asymmetric return shocks that are not captured by Gaussian innovations.

2.6 ARMA-GARCH model diagnostics

The adequacy of the *final* ARMA–GARCH specification is assessed using Ljung–Box tests on the standardised residuals and on their squares (Table 11).

Index	Q statistic	df	p-value
SPX	9.6667	10	0.4702
DAX	10.564	10	0.3924

Table 11: Ljung–Box tests on squared standardised residuals of the final ARMA–GARCH models.

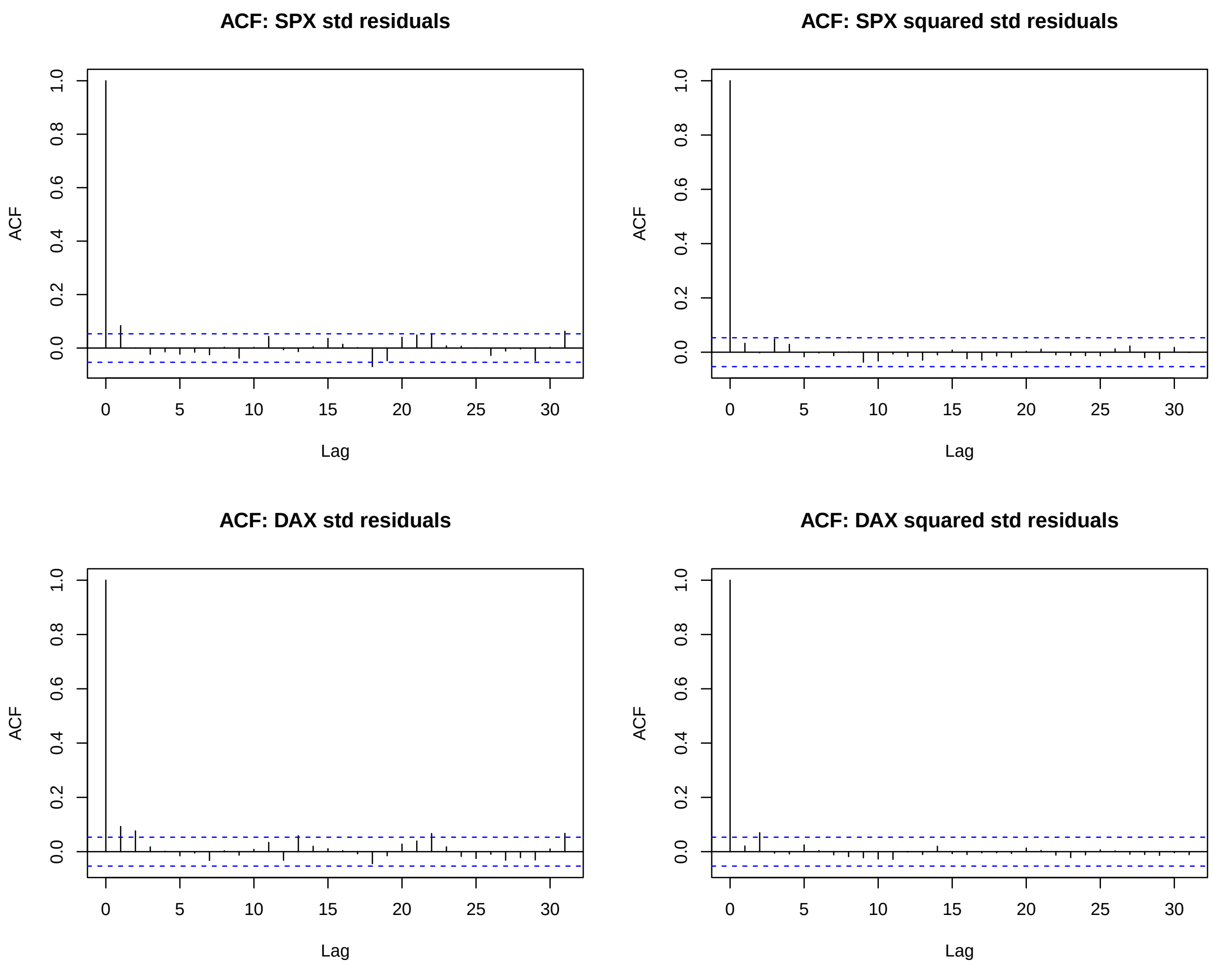

Figure 4: ACF of standardised residuals and standardised squared residuals for the final ARMA–GARCH models fitted to SPX and DAX returns.

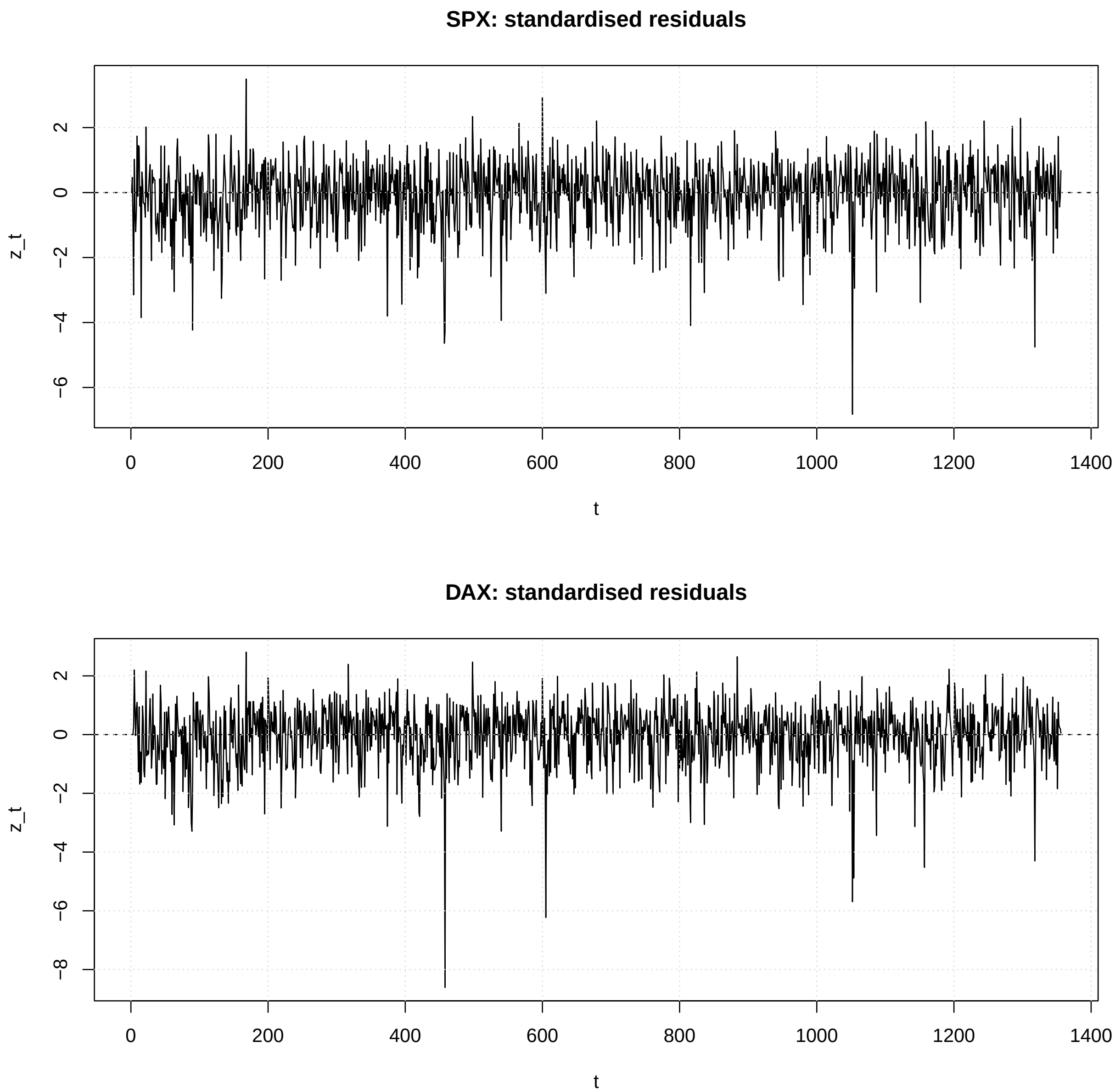

Figure 5: Time series of standardised residuals from the selected ARMA–GARCH models for SPX and DAX.

Although the ACFs of the standardised residuals in Figure 4 indicate minor residual autocorrelation at a few lags, there is no strong systematic pattern, suggesting that the ARMA component has captured most of the dependence in the conditional mean, in line with Tables 3 and 5. Likewise, the ACFs of the squared standardised residuals in Figure 4 show only limited remaining dependence, while the Ljung-Box results for the final model in Table 11 indicate no material residual autocorrelation in the squared residuals. Consistent with this, the time series of standardised residuals in Figure 5 are centred around zero and show no obvious remaining volatility clustering, indicating that the ARMA-GARCH specifications have largely filtered the conditional mean and variance dynamics. The resulting innovations are therefore approximately i.i.d., making them suitable for the probability integral transform. On this basis, we retain the ARMA-GARCH specification as the final model and proceed with the remaining diagnostic checks.

2.7 Probability integral transform & copula selection

Let $\hat{z}_{j,t}$ denote the fitted standardised residuals obtained from the fitted ARMA–GARCH marginal model for each series j , defined as:

$$\hat{\varepsilon}_{j,t} = r_{j,t} - \hat{\mu}_{j,t}, \qquad \hat{z}_{j,t} = \frac{\hat{\varepsilon}_{j,t}}{\hat{\sigma}_{j,t}}.$$

Since filtering is applied marginally, it removes serial dependence in the conditional mean and variance of each series independently, but does not eliminate cross-sectional dependence.
The PIT is applied under the fitted skewed students-t (sstd) distribution to map the standardised residuals to the unit interval:

$$u_{j,t} = \hat{F}_{\text{sstd}}\left(\hat{z}_{j,t};\, \hat{\nu}_j, \hat{\xi}_j\right),$$

where $\hat{\nu}_j$ and $\hat{\xi}_j$ are the fitted degrees of freedom and skewness parameters for series j. Under correct marginal specification, $u_{j,t} \sim$ i.i.d. $U(0,1)$. To avoid degeneracy at the boundary, the pseudo-observations are clamped to $[10^{-6}, 1 - 10^{-6}]$.

```r
#extract sstd condition/innovation
parspx <- model13@fit$par
pardax <- model23@fit$par

#extract the parameters from the distribution
nuspx <- unname(parspx["shape"])
xispx <- unname(parspx["skew"])
nudax <- unname(pardax["shape"])
xidax <- unname(pardax["skew"])
#apply the distribution to the standardized residuals
uspx <- psstd(z_spx, nu = nuspx, xi = xispx)
udax <- psstd(z_dax, nu = nudax, xi = xidax)

#clamp
eps <- 1e-6
uspx <- pmin(pmax(uspx, eps), 1 - eps)
udax <- pmin(pmax(udax, eps), 1 - eps)

#combine pairs to fit a copula
U <- na.omit(cbind(uspx, udax))
colnames(U) <- c("USPX","UDAX")
```

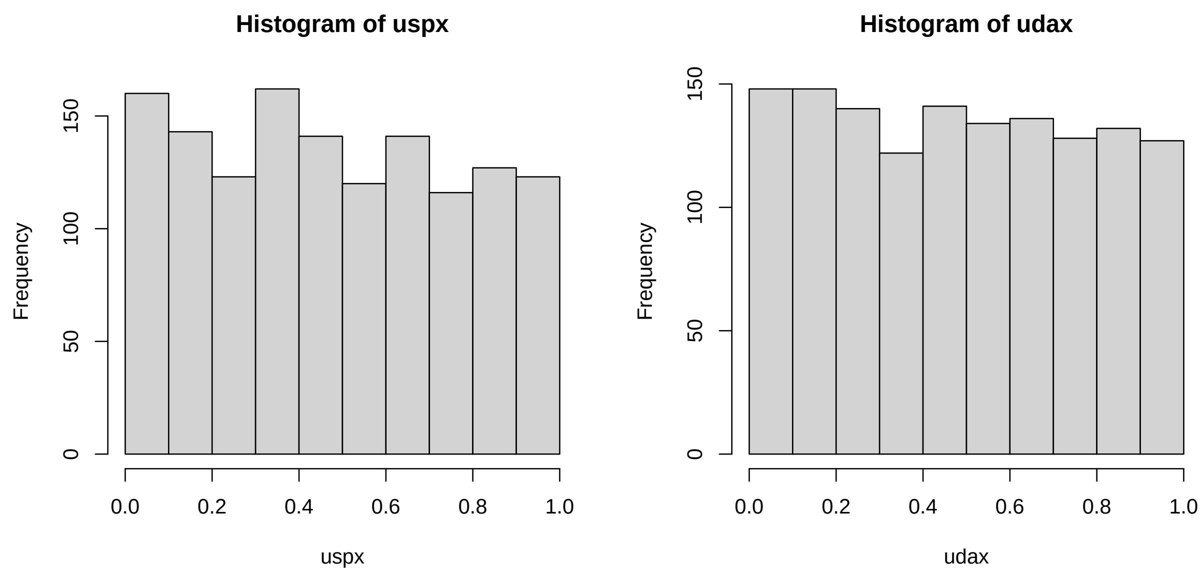

Figure 6: Pseudo-observations for SPX and DAX generated from the fitted skewed Student-t innovation distributions. The histograms are approximately uniform, therefore supporting their suitability for copula modelling.

Uniformity of the pseudo-observations is assessed via the Lilliefors-corrected Kolmogorov–Smirnov (KS) and Anderson–Darling tests.

Series	KS p-value	AD p-value
SPX	0.024	0.01216645
DAX	0.3472	0.1760574

Table 12: Kolmogorov–Smirnov and Anderson–Darling test p-values for pseudo-observations.

For DAX, neither test rejects uniformity at the 5% significance level (Table 12). For SPX, both tests reject normality, suggesting the data may not follow a uniform distribution (Table 12). Despite failing the tests, we decided to proceed regardless as uniformity tests with large samples can reject for small deviations that have limited impact on VaR; many copula workflows proceed if diagnostics still broadly look uniform and dependence fitting is stable(Blinov and Lemeshko 2014).

A bivariate copula is fitted via maximum likelihood on the pseudo-observations, with model selection via AIC, Kendall's τ, and tail dependence coefficients. The candidate copulas considered span over a broad set of families. Some common examples include:

- Gaussian(implicit): elliptical, no tail dependence
- Student-t (implicit): elliptical, symmetrical tail dependence
- Clayton (explicit): lower-tail dependence
- Gumbel (explicit): upper-tail dependence
- Frank (explicit): no tail dependence, flexible central dependence
- Survival BB1(implicit): asymmetric non-linear dependence in tails (two-parameter "Clayton– Gumbel" family, rotated via survival)

Historical VaR is included as a nonparametric benchmark, though being unconditional and backward-looking, it is regime-dependent and may be unreliable as a predictor in the presence of volatility clustering and structural change.

```r
model <- BiCopSelect(uspx, udax,
                     familyset = NA,
                     selectioncrit = "AIC",
                     indeptest = TRUE,
                     level = 0.05)

model
#Simulate N draws from the selected copula

N <- 1350
set.seed(1)

usim <- BiCopSim(N,
                 family = model$family,
                 par    = model$par,
                 par2   = model$par2)
```

The selected copula is Survival BB1 structure, implying asymmetrical tail dependence:

$$\hat{C} = C_{\text{sBB1}}\left(u_1, u_2; \hat{\theta}, \hat{\delta}\right), \qquad \hat{\theta} = 0.24,\ \hat{\delta} = 1.85, \qquad \hat{\tau} = 0.52.$$

As a qualitative validation, $N = 1350$ pairs are simulated from $\hat{C}$ and compared visually against the observed PIT pairs to confirm that the fitted copula reproduces the empirical dependence pattern (Figure 7).

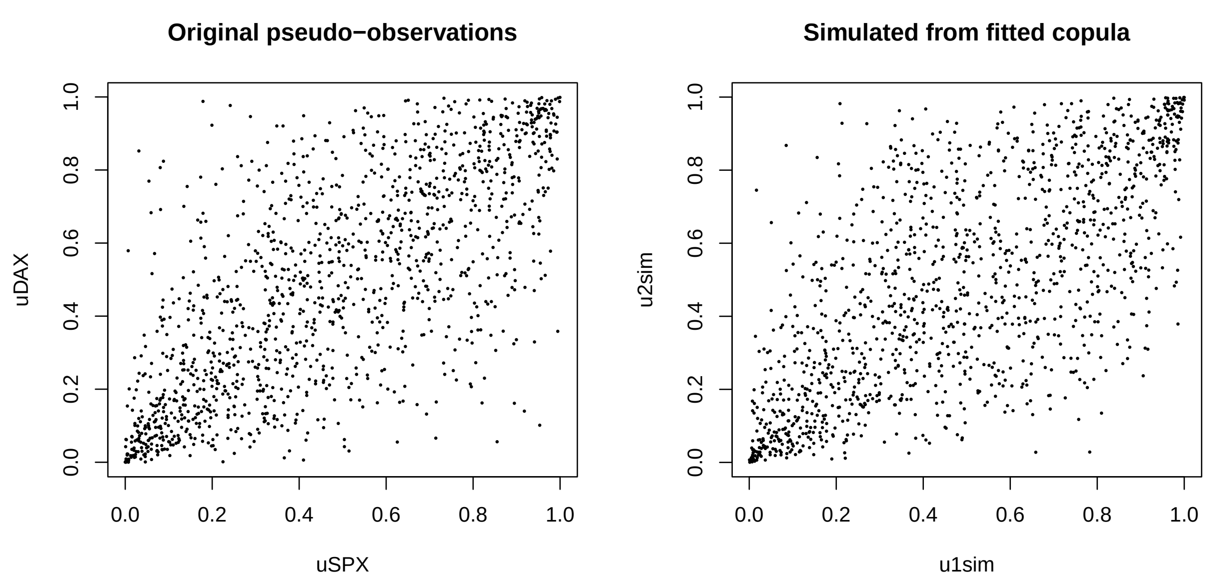

Figure 7: Scatterplots of the original pseudo-observations and pseudo-observations simulated from the fitted Survival BB1 copula. The simulated sample visually matches the empirical dependence structure, supporting the adequacy of the selected copula model.

2.8 VaR estimation

Let $(U^{(i)}_{\text{SPX}}, U^{(i)}_{\text{DAX}})$ be $N_{sim} = 50,000$ simulated draws from the fitted copula $\hat{C}$, for $i = 1, 2, \dots, N_{sim}$ Monte Carlo scenarios. Each draw is mapped back to the innovation scale via the inverse fitted marginal CDF,

$$Z^{(i)}_j = \hat{F}^{-1}_{Z_j}\left(U^{(i)}_j\right), \qquad j \in \{\text{SPX}, \text{DAX}\},$$

the one-step-ahead return for scenario i is reconstructed as

$$r^{(i)}_{j,T+1} = \hat{\mu}_{j,T+1} + \hat{\sigma}_{j,T+1}\, Z^{(i)}_j,$$

Where $\hat{\mu}_{j,T+1}$ and $\hat{\sigma}_{j,T+1}$ are the ARMA–GARCH predicted conditional mean and volatility $T+1$. Portfolio returns are then formed as an equal-weight combination of $r^{(i)}_{\text{SPX},T+1}$ and $r^{(i)}_{\text{DAX},T+1}$, and VaR is estimated as the corresponding empirical quantile.

To assess the sensitivity to the choice of dependence structure, VaR estimates are computed under the Survival BB1 copula and five alternatives (Gaussian, Student-t, Clayton, Gumbel, and Frank), alongside a historical empirical benchmark.

```r
set.seed(1)
Nsim <- 50000

# last-step mu/sigma (one-week ahead demo)
mu1  <- as.numeric(tail(fitted(model13), 1)); sig1 <- as.numeric(tail(volatility(model13), 1))
mu2  <- as.numeric(tail(fitted(model23), 1)); sig2 <- as.numeric(tail(volatility(model23), 1))

# fit common copulas on your PIT data (reuses uspx, udax)
fit_gauss   <- BiCopEst(uspx, udax, family=1)
fit_t       <- BiCopEst(uspx, udax, family=2)
fit_clayton <- BiCopEst(uspx, udax, family=3)
fit_gumbel  <- BiCopEst(uspx, udax, family=4)
fit_frank   <- BiCopEst(uspx, udax, family=5)

# simulate U for each copula
U_sel     <- BiCopSim(Nsim, family=model$family, par=model$par, par2=model$par2)
U_gauss   <- BiCopSim(Nsim, family=1, par=fit_gauss$par)
U_t       <- BiCopSim(Nsim, family=2, par=fit_t$par, par2=fit_t$par2)
U_clayton <- BiCopSim(Nsim, family=3, par=fit_clayton$par)
U_gumbel  <- BiCopSim(Nsim, family=4, par=fit_gumbel$par)
U_frank   <- BiCopSim(Nsim, family=5, par=fit_frank$par)

# VaR from U sample
VaR_from_U <- function(U){
  z1 <- qsstd(U[,1], nu=nuspx, xi=xispx)
  z2 <- qsstd(U[,2], nu=nudax, xi=xidax)
  r1 <- mu1 + sig1*z1
  r2 <- mu2 + sig2*z2
  rP <- log(1+0.5*(exp(r1)-1) + 0.5*(exp(r2)-1))
  quantile(rP, c(0.01, 0.05))
}

VaR_sel     <- VaR_from_U(U_sel)
VaR_gauss   <- VaR_from_U(U_gauss)
```

```r
VaR_t       <- VaR_from_U(U_t)
VaR_clayton <- VaR_from_U(U_clayton)
VaR_gumbel  <- VaR_from_U(U_gumbel)
VaR_frank   <- VaR_from_U(U_frank)

VaR_hist <- quantile(returns$rP_rebal, c(0.01, 0.05), na.rm=TRUE)

table <- rbind(
  setNames(as.data.frame(t(VaR_sel)),     c("1%","5%")),
  setNames(as.data.frame(t(VaR_gauss)),   c("1%","5%")),
  setNames(as.data.frame(t(VaR_t)),       c("1%","5%")),
  setNames(as.data.frame(t(VaR_clayton)), c("1%","5%")),
  setNames(as.data.frame(t(VaR_gumbel)),  c("1%","5%")),
  setNames(as.data.frame(t(VaR_frank)),   c("1%","5%")),
  setNames(as.data.frame(t(VaR_hist)),    c("1%","5%"))
)

row.names(table) <- c(paste0("Selected: ", model$familyname),
                     "Gaussian","Student_t","Clayton","Gumbel","Frank","Historical")
table
```

Copula	$\text{VaR}_{0.01}$	$\text{VaR}_{0.05}$
Selected: Survival BB1	-0.04979521	-0.02890508
Gaussian	-0.04829552	-0.02886399
Student-t	-0.04937962	-0.02863751
Clayton	-0.05124481	-0.02983666
Gumbel	-0.04421892	-0.02757014
Frank	-0.04366806	-0.02874826
Historical	-0.07400532	-0.04120645

Table 13: Value-at-Risk estimates at the 1% and 5% levels for the selected Survival BB1 copula, the alternative copula models, and the historical benchmark.

Under the Survival BB1 copula, $VaR_{0.95} = 0.0289$ and $VaR_{0.99} = 0.0498$ (Table 13). Alternative copulas yield broadly consistent estimates; the Clayton copula produces the most conservative model-based estimate ($Q_{0.01} = -0.0512$), consistent with its lower tail dependence, whilst the Frank and Gumbel copulas produce the least conservative ($Q_{0.01} = -0.0437$ and -0.0442 respectively).

The historical benchmark is substantially more conservative ($VaR_{0.95} = 0.0412, VaR_{0.99} = 0.0740$), as it pools realised returns conditionally across the full 2000-2025 sample, including crisis periods, rather than conditioning on the current volatility state $\hat{\sigma}_{j,T+1}$.

2.9 Model evaluation

Backtesting evaluates VaR model adequacy by comparing predicted loss thresholds against realised returns. The Kupiec test assesses whether the observed exceedance rate is consistent with the nominal coverage level, whilst the Christoffersen test additionally examines whether exceedances are independently distributed over time, detecting clustered violations that would indicate a failure to capture volatility dynamics. VaR forecasts at $\alpha = 0.05$ and $\alpha = 0.01$ are backtested over $T = 1355$ weekly observations.

```r
set.seed(1)
Nsim <- 50000
U <- BiCopSim(Nsim, family=model$family, par=model$par, par2=model$par2)

# sstd shocks
p1 <- model13@fit$par; p2 <- model23@fit$par
z1 <- qsstd(U[,1], nu=unname(p1["shape"]), xi=unname(p1["skew"]))
z2 <- qsstd(U[,2], nu=unname(p2["shape"]), xi=unname(p2["skew"]))

# 1-step paths + realised portfolio return
mu1 <- as.numeric(fitted(model13)); s1 <- as.numeric(volatility(model13))
mu2 <- as.numeric(fitted(model23)); s2 <- as.numeric(volatility(model23))
Tn <- min(length(mu1), length(mu2), nrow(returns))

rP_real <- returns$rP_rebal[2:Tn]

# iterate over weeks calculating VaR
VaRret <- function(q){
  v <- numeric(Tn-1)
  for(t in 1:(Tn-1)){
    r1 <- mu1[t] + s1[t]*z1
    r2 <- mu2[t] + s2[t]*z2
    rP <- log(1 + 0.5*(exp(r1)-1) + 0.5*(exp(r2)-1))
    v[t] <- as.numeric(quantile(rP, probs=q))
  }
  v
}

VaR_95 <- VaRret(0.05)
VaR_99 <- VaRret(0.01)

vt95 <- VaRTest(alpha=0.05, actual=rP_real, VaR=VaR_95)
vt99 <- VaRTest(alpha=0.01, actual=rP_real, VaR=VaR_99)

data.frame(
  level=c("95%","99%"),
  T=c(length(rP_real), length(rP_real)),
  expected_exceed=c(vt95$expected.exceed, vt99$expected.exceed),
  observed_exceed=c(vt95$actual.exceed, vt99$actual.exceed),
  exceed_rate=c(vt95$actual.exceed/length(rP_real), vt99$actual.exceed/length(rP_real)),
  Kupiec_p=c(vt95$uc.LRp, vt99$uc.LRp),
  Christoffersen_p=c(vt95$cc.LRp, vt99$cc.LRp)
)
```

Level	T	Expected Exceedances	Observed Exceedances	Exceedance Rate	Kupiec p-value	Christoffersen p-value
95%	1355	67	77	0.05682657	0.2587006	0.06020675
99%	1355	13	16	0.01180812	0.5154702	0.32996970

Table 14: Backtesting of 1-day-ahead VaR forecasts.

At the 95% level, 77 exceedances were observed against 67 expected ($rate = 0.0568$), with the Kupiec and Christoffersen tests returning p-values of 0.259 and 0.060 respectively, neither rejecting at the 5% level (Table 14). At the 99% level, 16 exceedances were observed against 13 expected ($rate = 0.0118$), with p-values of 0.515 and 0.330 respectively. At both confidence levels, the model produces VaR forecasts consistent with empirical exceedance frequency and independence.

2.10 Limitations

Several limitations warrant acknowledgement. Uniformity diagnostics indicate residual departure from $Uniform(0, 1)$ for the SPX pseudo-observations, suggesting marginal misspecification that may affect copula estimation and tail inference. Model-based VaR estimates are also considerably less conservative than historical empirical quantiles, reflecting the fact that the model conditions on current GARCH volatility rather than the unconditional return distribution across all regimes. Finally, copula selection relied on AIC and visual diagnostics, no formal goodness-of-fit test was conducted, representing a natural avenue for further refinement.

3. Conclusion

Using weekly SPX and DAX returns over $2000 - 2025$, ARMA–GARCH marginal models with skewed Student-t innovations were fitted to each series, capturing volatility clustering, heavy tails, and serial dependence. Following the PIT transformation, AIC selected a Survival BB1 copula with $(\hat{\theta}, \hat{\delta}) = (0.24, 1.85)$ and Kendall's $\tau = 0.52$, reflecting strong asymmetric tail dependence between the two indices. Monte Carlo simulation ($N = 50,000$) yielded $VaR_{0.95} = 0.0289$ and $VaR_{0.99} = 0.0498$ for an equal-weight rebalanced portfolio, with Kupiec and Christoffersen backtests confirming exceedance frequency and independence consistent with nominal coverage at both levels.

4. References

Blinov, Pavel, and Boris Lemeshko. 2014. “A Review of the Properties of Tests for Uniformity.” In *2014 12th International Conference on Actual Problems of Electronic Instrument Engineering, APEIE 2014 - Proceedings*. https://doi.org/10.1109/APEIE.2014.7040743.

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